

UDITA SEN

+91 9475318579 | uditasen.makaut.it@gmail.com | [LinkedIn](#) | [LeetCode](#)

Bengaluru, India · Open to relocation

PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience — 4+ at PayPal — owning REST API design, distributed systems, and real-time data infrastructure on GCP. Specializes in high-throughput streaming pipelines, fault-tolerant service design with idempotency and backpressure, and database-layer optimization across systems processing billions of records daily.

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL

Backend & APIs: Spring Boot, REST API Design, Microservices, gRPC

Streaming & Pipelines: Apache Kafka, GCP Pub/Sub, Dataflow, Apache Beam, Spark, Airflow, Amazon Kinesis

Databases & Caching: BigQuery, Bigtable, MySQL, PostgreSQL, Redis, Elasticsearch

Cloud & Infra: GCP, Kubernetes, Jenkins CI/CD, Docker

Observability: Datadog, Kafka consumer lag tracking, production alerting, centralized logging

Concepts: Distributed Systems, Event-Driven Architecture, Schema Evolution, LLD & HLD, Real-Time ETL

EXPERIENCE

PayPal

Bangalore, India

Software Engineer 2 — Data Foundation Platform

Jul 2024 – Present

- Architected a multi-step event-driven data platform (7.5B+ records/day): Pub/Sub ingestion, pluggable producer/consumer components, schema validation, transformation, and multi-sink fan-out; each pipeline runs in an isolated runtime environment, onboarding new streams through reusable shared components.
- Migrated 500K+ daily batch ETL jobs to near-real-time streaming (Java, Pub/Sub, Dataflow) with idempotent consumers and dead-letter handling; reduced end-to-end latency from 6 hrs to 10 min for fraud and analytics teams.
- Designed a bidirectional REST layer (BigQuery ↔ HTTP, 2B+ calls/day) with exponential backoff, schema validation, and circuit breakers; handles inbound ingestion and outbound delivery without data loss.
- Unified 200M+ customer profiles across three payment platforms via probabilistic identity resolution and schema normalization; golden records in BigQuery feed risk, fraud, and ML consumers.
- Engineered a BigQuery → Bigtable sync service using an orchestrated worker architecture over Pub/Sub — 53M records in 20 min; chunk-level retry and replayable recovery on partial failure.
- Shipped a GenAI ops agent (LangChain) for anomaly detection over streaming tables; cut analyst investigation time by 40% on recurring incident patterns.

Software Engineer 1 — Enterprise Data Infrastructure

Jan 2022 – Jul 2024

- Re-platformed 800TB+ data infrastructure from an on-premises Hive platform to GCP; maintained 200+ production Spark and Pub/Sub pipelines through a zero-downtime migration.
- Redesigned a large-scale data warehouse on BigQuery with query-pattern-aligned partitioning and clustering; cut reporting costs 60% (~\$4M/year) and improved p99 query latency by 3x.
- Replaced Informatica workflows with Spark + Oracle GoldenGate CDC; enabled real-time Oracle → BigQuery ingestion at 7.6B+ records/day with schema drift detection for major merchant partners.
- Modernized Risk-Based Lending from batch to real-time streaming pipelines; improved data freshness SLA across 50+ downstream risk scoring and ML systems.

HashedIn by Deloitte

Bangalore, India

Software Engineer — Internal Platform Team

Jul 2021 – Dec 2021

- Built RESTful APIs (Java, Spring Boot) and Angular frontend for an internal LMS — course management, enrollment, RBAC, and admin flows; converted from intern to full-time engineer within 3 months.

Innovation Lab, MAKAUT

India

Researcher

Jan 2018 – Jan 2021

- Built ML models (Random Forest, XGBoost, SVM) predicting epidemic risk zones from remote-sensing data; co-authored findings with Prof. Somdatta Chakravorty.

EDUCATION

Maulana Abul Kalam Azad University of Technology, West Bengal

Jun 2017 – Jul 2021

B.Tech, Information Technology · CGPA: 8.63 / 10.0 · Department Rank: 2

ACHIEVEMENTS

- NPTEL Silver Medal — Database Management Systems (IIT Kharagpur), ranked 43rd / 743 · NPTEL Silver Medal — Algorithms (IIT Madras), ranked 68th / 1890
- 800+ problems solved across LeetCode, CodeChef, and HackerRank.